# CLIMB at VBS 2027: Multi-Signal Video Retrieval with Team-Based Collaboration

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**Abstract.** We present CLIMB, an interactive video retrieval system built for the Video Browser Showdown 2027. CLIMB retrieves information across four complementary signals: visual embeddings, on-screen text, generated captions, and speech transcripts. By combining rank-based fusion with a hybrid of lexical and semantic matching, the system ensures broad retrieval coverage. Additionally, queries can be chained into temporal sequences, providing increased flexibility while sharply narrowing the result set. We treat retrieval as a shared team activity, allowing operators to send hints to each other and automatically hiding teammate-submitted scenes for ad-hoc video search. Crucially, core functions such as search, browsing, and submission remain fully operational even if the shared collaboration service becomes unreachable.

**Keywords:** Video Browser Showdown · Interactive Video Retrieval · Collaborative Search · Semantic Embedding.

## 1 Introduction

The Video Browser Showdown (VBS) [5, 6, 11] is an interactive video retrieval competition that asks participants to solve known-item search (KIS), ad-hoc video search (AVS), and visual question answering (VQA/QAS) tasks against large-scale video collections. For VBS 2027, the main collection is V3C [12] with 28,450 videos, roughly 3,800 hours, and about 8.7 TB of full-resolution video data, along with the smaller collections MVK [17] and GynSurgLHE (LHE) [8].

Successfully completing the search tasks presents multiple fundamental challenges. A single visual embedding is often too vague to capture precise task hints, such as specific on-screen text or spoken dialog, and for difficult tasks, teammates have no direct way to point each other to promising results. Additionally, AVS scoring rewards distinct correct shots, but a team whose operators search independently spends a substantial share of its submissions re-finding what a teammate has already sent.

To address these challenges, CLIMB indexes videos as scenes and keyframes, retrieves over four independent signals, and fuses them by rank. It also lets operators send clues to their teammates and treats an AVS task as a single shared activity, hiding duplicate scenes from everybody's list.

Concretely, we contribute:

1. Keyframe-level retrieval fusing four heterogeneous signals by rank (Sect. 4.1).
2. Temporal chaining, `A >> B >>(120) C`, with a full query per stage (Sect. 4.3).
3. Network-resilient team collaboration: shared AVS state and hints (Sect. 6).
4. An index engineered for commodity hardware: binary-quantized HNSW with oversample-and-rerank (Sect. 3).

## 2 System Overview

CLIMB is built from six components (see Fig. 1). Decoding each video, segmenting it into scenes, selecting keyframes, computing embeddings, running OCR, as well as captioning and transcribing, happen in an offline pipeline described in Sect. 3. The resulting scenes, keyframes, and four indexes that the retrievers query land in a PostgreSQL database.

At query time, the system runs as two processes, a Node.js backend transmitting media to the frontend, handling DRES [10, 14] communication and collaboration features (Sect. 6), as well as a Python engine handling models, query embedding, and rank fusion. Both connect to the database: the backend reads metadata directly, delegating only search execution to Python. The frontend (Sect. 5) communicates exclusively with the Node.js backend.

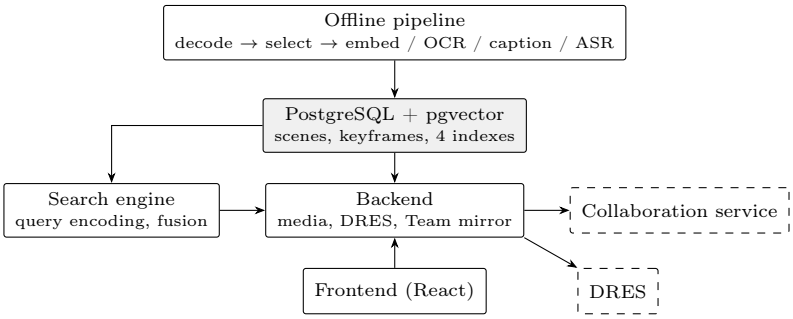



**Fig. 1.** CLIMB architecture. All components can run locally on each machine, except for two external services: DRES and the collaboration service, which is deployed once.

## 3 Indexing Pipeline

### 3.1 Scenes and Keyframe Selection

The indexing pipeline starts from the scene boundaries shipped with each dataset, produced with TransNetV2 [16] for V3C and with a keyframe extraction algorithm [15] based on ORB keypoint tracking [13] for LHE [8] and MVK [17]. MVK

clips from the same dive are shipped as separate clips, but they are grouped into one recording on a shared timeline, so exclusion, temporal chains and AVS coverage work per dive, while DRES submissions still reference the individual clips.

Within each scene, candidate frames are decoded at 2 fps, and the scene receives one keyframe per four seconds of duration, at least two and at most 32. Fades and blank frames (very dark, very bright, or low-contrast) are discarded first, unless they contain edges, so that title cards and credits remain available for OCR. Each remaining candidate is described by a cheap, model-free fingerprint: a color histogram, a perceptual hash, and the edge density. Selection follows a greedy farthest-point strategy [3], seeded with the frame closest to the scene's mean fingerprint.

### 3.2 Feature Extraction

CLIMB extracts four signals, two per keyframe and two per scene. Each keyframe is embedded into a 1024-dimensional space by SigLIP2 [18] and searched for on-screen text by PP-OCR [21]. Since surgical footage lies far from SigLIP2's training data, LHE keyframes are additionally embedded with PeskaVLP [20].

The two scene signals are textual and embedded into a 1024-dimensional space by multilingual-e5-large [19]. Given its cost and intra-scene redundancy, Qwen3-VL-4B-Instruct [1] captions only each scene's first keyframe. Audio transcription is handled by Whisper large-v3 [9]. Transcriptions produced from music or noise, identified as repetitive or isolated from other speech, are excluded.

Embeddings are indexed with HNSW [7] over binary-quantized vectors, and candidates are reranked against their half-precision vectors. Quantization shrinks the index for 12.4M keyframes from 28 GB at half precision to 5.3 GB.

## 4 Retrieval Model

### 4.1 Rank-Based Fusion at Keyframe Level

Scores from four different retrievers are not comparable, so CLIMB fuses the signals by rank, avoiding model-specific calibration. Visual and OCR signals rank keyframes directly, while the caption and speech signals rank scenes, whose keyframes inherit the scene's rank. All lists are then merged via weighted Reciprocal Rank Fusion (RRF) [2], which lets some signals count more than others. The fused score of a keyframe $d \in D$ over a set of retrievers $R$ is

$$RRFscore(d \in D) = \sum_{r \in R} \frac{w(r)}{60 + r(d)}, \tag{1}$$

where $r(d)$ is the rank assigned by encoder $r \in R$ with weight $w(r)$. The constant 60, empirically determined in [2], reduces the dominance of top-ranked keyframes. For V3C, the uploader's title, description and tags are also matched against the query as a video-level prior: each retrieved keyframe receives an extra RRF term for its video's rank. The system's default weights are 1 for visual,

caption, speech and metadata, 4 for OCR and 8 for explicit matches via the `said:` or `text:` operator (Sect. 4.3).

### 4.2 Semantic and Lexical Matching

Although cross-modal visual retrieval with SigLIP2 is inherently semantic, the remaining text-to-text signals could rely on lexical matching. For generated captions, however, this fails on vocabulary mismatch: a query for "law enforcement" cannot match a caption describing "a police uniform", and neither "law" nor "enforcement" appears in any of our captions.

ASR transcripts are also embedded semantically to allow for conceptual and cross-lingual search, but users can invoke a `said:` operator for exact lexical matching of specific quotes. In contrast, OCR operates as a purely lexical signal to pinpoint exact on-screen text provided in task hints.

### 4.3 Query Language and Temporal Chains

The `said:` operator is one of several prefixes (Table 1) that narrow down a plain text query over all four signals. Each restricts the search to the videos that satisfy it, ranking the plain text only within them. Once the prefixes leave at most 300 videos, they are searched exhaustively, bypassing the vector index.

For tasks describing an ordered sequence of scenes, `>>` joins queries into a temporal chain. Each stage is a complete query including its own prefixes, so `text: pegasus >> mountain bike race` finds a scene showing the text "pegasus", followed by one that matches a mountain bike race. Consecutive stages must lie in the same video and are by default at most 60 seconds apart. Writing `>>(120)` raises this to 120 seconds. The maximum is ten minutes and four stages.

Each stage returns a ranked list of keyframes, grouped by video. Videos lacking a keyframe for any stage are excluded. For every first-stage hit, the best continuation is found by working backwards from the last stage, so each pair of adjacent stages is compared only once. At most three chains are kept per video, and all chains are ranked by their score, the sum of their stages' RRF scores.

**Table 1.** Query syntax. Prefixes are video-level requirements. They are repeatable and can be combined. `>>` chains queries temporally within a single video.

| Input | Effect |
|---|---|
| `a drawing of people on a beach` | Global search (all signals) |
| `text: state capitol` | Scenes showing this text |
| `said: new york` | Scenes in which this is said |
| `tag: travel` | Only videos tagged with `travel` |
| `video:00191 video:00192` | Only these videos |
| `-video:00191` | Exclude this video |
| `a bridge >> beach >> sunset` | Sequential search (same video) |

## 5 Interactive Search Interface

The CLIMB user interface is shown in Fig. 2. A search typically follows an iterative loop. The operator selects a collection (V3C, MVK, or LHE) and enters a query, which returns a grid of keyframes. Selecting one plays a five-second scene preview from its timestamp, while the filmstrip below the player shows all keyframes for quick navigation. The submission time is prefilled from the keyframe but stays editable for fine-tuning. *Submit to DRES* sends the selection to the evaluation server, *Exclude video from search* removes a video from the results, and when textual queries fall short, *Find Similar* enables visual query-by-example. For VQA tasks, operators type the answer below the filmstrip and submit it as text alone or together with the selected shot. Results are cached, so scrolling loads further pages without re-running the search.

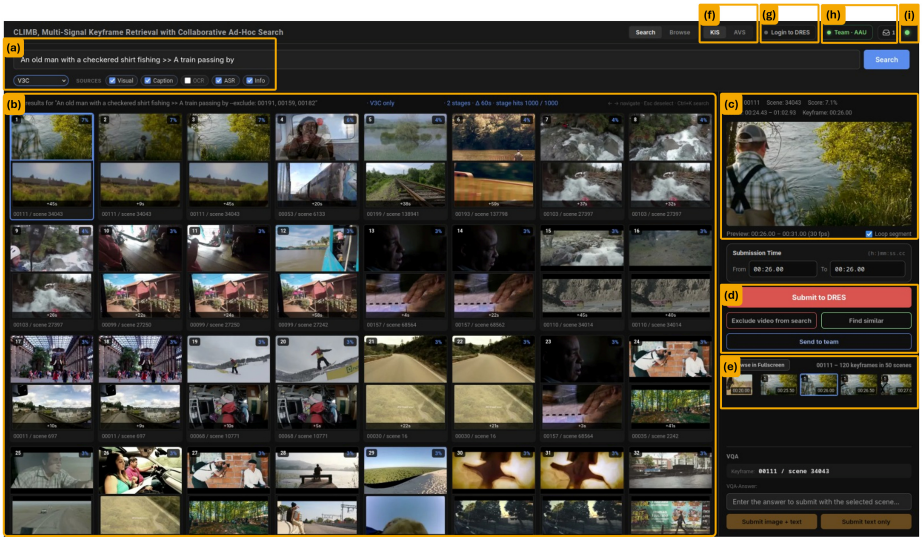



**Fig. 2.** The CLIMB interface: (a) query bar, source filters and dataset selection, (b) scene result grid, (c) detail panel and video player, (d) submission and search-refinement controls, (e) keyframe filmstrip, (f) AVS session tab, (g) DRES connection controls, (h) team connection controls, and (i) system health status.

## 6 Team Collaboration

At VBS, every operator runs a complete local CLIMB instance, avoiding network dependency for media delivery. Nonetheless, allowing operators to work together is highly beneficial, as teammates can share promising leads, reducing wrong or duplicate submissions while speeding up the search.

To achieve this, operators can send a teammate a hint: a keyframe or an entire video, along with an optional note. The hint appears as a notification, which, when clicked, opens a browsing view of all keyframes with the sent keyframe selected. Whether opened or dismissed, it is kept in an inbox, so that operators can come back to it later.

For AVS, collaboration goes further. Because AVS rewards distinct correct shots, a team in CLIMB shares one AVS session. Submitted scenes are hidden from the results of all team members. Other scenes of these videos are marked as covered, steering the team towards new videos, but stay visible, since one video can hold several correct shots.

This shared state is maintained by a lightweight collaboration service, deployed once. Operators join a team by name and password, with each backend keeping a local copy of its team's state (the current AVS session and all hints), refreshed every four seconds. Importantly, searches and submissions must not depend on the speed or availability of the collaboration service. Searches therefore read only the local copy, and submissions go to DRES first and are recorded with the service afterwards, so a slow service never delays them. If the service becomes unreachable, the local copy is kept rather than cleared, so the team's exclusions survive an unstable connection.

## 7 Preliminary Results

A far simpler CLIMB prototype won our university's course-internal retrieval competition, but VBS is a much greater challenge. CLIMB is implemented for the full V3C collection, yet the V3C metrics below were measured on V3C1 (7,471 videos, 999 h, 521,573 scenes, 1,349,785 keyframes).

Compared with exhaustive exact search (2.4 s), oversampling 2,000 candidates before exact reranking recovers on average 88.7% of the true top 20 in 80 ms, 4,000 candidates recovers 93.8% in 159 ms, and 8,000 97.1% in 309 ms. CLIMB defaults to 4,000 candidates, while keeping this parameter configurable.

Since neither MVK nor LHE resembles the imagery SigLIP2 was trained on, we tested domain-specific models on both. Fusing PeskaVLP with SigLIP2 raises the relevant share of the first LHE result page from 46% to 54%, whereas BioCLIP 2.5 [4] alone lowers the share of MVK queries with a relevant frame on the first page from 93% to 64%, so it is not used.

## 8 Conclusion

In this paper, we present CLIMB, an interactive video retrieval system for VBS 2027. It uses rank-based fusion to combine visual embeddings, on-screen text, generated captions, and transcripts at the keyframe level and supports temporally chained queries. A collaborative mode hides teammate-submitted scenes and degrades gracefully, keeping core search functions fully available offline.

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